Kinetic Turbulence, Heating, and Particle Acceleration

Advanced Topics in Plasma Physics

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These notes follow the standard kinetic-cascade framework for strongly magnetized, weakly collisional plasmas, then use exact energy budgets and published kinetic-simulation results to organize particle heating, electron–ion energy partition, and nonthermal acceleration. The aim is a usable physical map rather than a derivation of a complete gyrokinetic or particle-in-cell model.

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1 The question left by fluid turbulence

MHD turbulence explains how energy moves from a large injection scale toward smaller spatial scales. It does not by itself determine what happens when the cascade reaches ion and electron kinetic scales. At that point the central questions change:

- Which fluctuations carry the cascade below the ion scale?
- How is field energy transferred to ions and electrons?
- Which part becomes thermal energy, and which part forms a nonthermal tail?
- Which reduced description—gyrokinetics, hybrid kinetics, or full PIC—retains the required physics?

The organizing framework in this chapter is the one developed for astrophysical gyrokinetics [1]: energy injected at fluid scales can cascade in both physical space and velocity space. The spatial cascade reaches ion and electron scales, while phase mixing creates increasingly fine velocity-space structure. Weak collisions, however small, can act on sufficiently fine structure and complete the conversion to heat.

Takeaway. A spectral cascade, particle energization, and irreversible heating are related but distinct statements. A complete argument must connect them with an energy or entropy budget.

2 From an MHD cascade to kinetic scales

Let L be the outer scale and let ρ_i and ρ_e denote characteristic ion and electron gyroradii. A useful first map is

$$L \gg \rho_i \gg \rho_e.$$

For $k_\perp \rho_i \ll 1$, low-frequency Alfvénic fluctuations are well described by an MHD-like cascade. Near $k_\perp \rho_i \sim 1$, finite-Larmor-radius and kinetic effects become unavoidable. Under the low-frequency, anisotropic ordering of gyrokinetics, the Alfvénic cascade continues as a kinetic-Alfvén-wave cascade, while ion-entropy fluctuations can undergo a separate phase-space cascade [1]. Near electron scales, electron kinetics determines the remaining conversion pathways.

The often-quoted $k_\perp^{-5/3}$ and sub-ion slopes are phenomenological summaries, not universal laws. The observed or simulated slope can depend on plasma beta, imbalance, intermittency, damping, driving, and available scale separation. For this lecture, the important point is that the kinetic range provides several competing channels for transferring energy out of the fluctuating fields.

2.1 What a spectral break does and does not establish

A steepened spectrum near $k_\perp \rho_i \sim 1$ shows that the cascade rate, wave physics, or both have changed. It does not alone show

- which species gained the missing energy;
- whether the gain is in bulk motion, a thermal core, or a nonthermal tail;
- whether the energy exchange is reversible over a longer interval; or
- whether a local drop in field energy left the system through a boundary.

Those questions require the budgets in the next section.

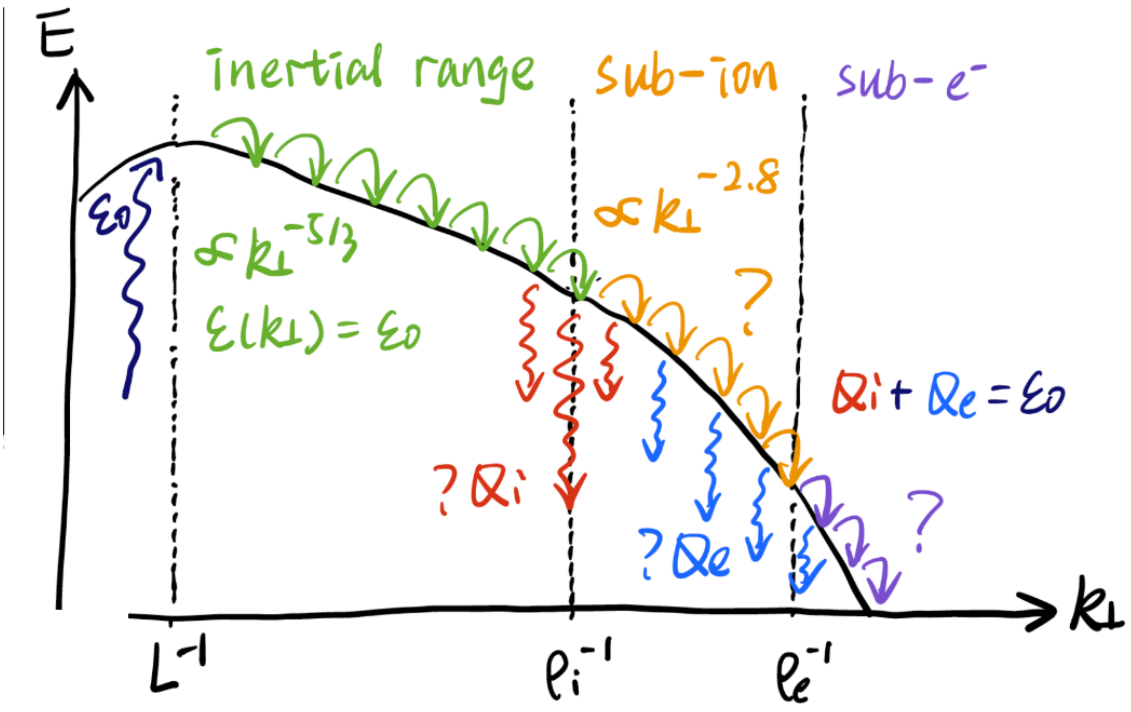


Figure 1: A scale-by-scale picture of the cascade and the unresolved partition of the injected power into ion and electron internal energy. The spectral break identifies a change of physics; it does not by itself measure a heating rate.

3 The energy budget: what is actually being measured?

For a nonrelativistic Vlasov–Maxwell plasma, define the electromagnetic energy density

$$W_{\mathrm{EM}} = \frac{\epsilon_0 E^2}{2} + \frac{B^2}{2\mu_0}.$$

Poynting's theorem is

$$\frac{\partial W_{\mathrm{EM}}}{\partial t} + \nabla\cdot \boldsymbol{S} = -\sum_s \boldsymbol{j}_s\cdot\boldsymbol{E}. \tag{1}$$

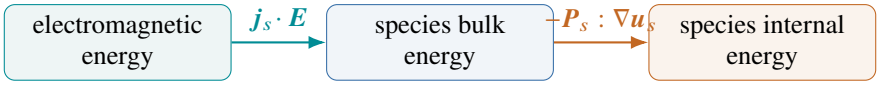
Thus $\boldsymbol{j}_s\cdot\boldsymbol{E}$ measures transfer between the fields and the total particle energy of species s.

For each species, separate bulk kinetic energy $K_s = \rho_s u_s^2/2$ from internal energy $U_s = \mathrm{tr}\,\boldsymbol{P}_s/2$. Their moment equations can be written

$$\frac{\partial K_s}{\partial t} + \nabla\cdot(K_s\boldsymbol{u}_s + \boldsymbol{P}_s\cdot\boldsymbol{u}_s) = \boldsymbol{j}_s\cdot\boldsymbol{E} + \boldsymbol{P}_s : \nabla\boldsymbol{u}_s, \tag{2}$$

$$\frac{\partial U_s}{\partial t} + \nabla\cdot(U_s\boldsymbol{u}_s + \boldsymbol{q}_s) = -\boldsymbol{P}_s : \nabla\boldsymbol{u}_s. \tag{3}$$

The same terms appear with opposite signs, so the sum of field, bulk, and internal energies closes after boundary fluxes, forcing, radiation, and collisions are included. The physical chain is



This diagram is schematic: $j_s \cdot E$ can also change random particle energy directly, and the pressure–strain term contains compressive and incompressive contributions. Equations (1)–(3), rather than the arrows alone, define the accounting.

Diagnostic	What it establishes	What it does not establish by itself
$j_s \cdot E$	Field–particle energy exchange for species s	Thermal versus nonthermal gain; irreversibility
$-P_s : \nabla u_s$	Bulk-to-internal conversion	Whether the conversion is reversible over a cycle
$f_s(x, v)$	Where the distribution changes	A unique energization mechanism from the final shape alone
Entropy or coarse-grained entropy	Loss of resolved phase-space information	A mechanism unless resolution and collisions are specified

Table 1: Each diagnostic answers a different question.

4 Energization, heating, and acceleration

These words should be kept separate.

Energization means an increase of particle energy. It may include bulk acceleration, reversible compression, thermal broadening, or formation of a tail.

Heating means an increase of internal energy, with irreversible heating requiring an entropy-producing or explicitly coarse-grained endpoint. In an ideal Vlasov system, phase mixing can hide energy in fine velocity-space structure while the fine-grained entropy remains conserved.

Nonthermal acceleration means that a part of the distribution develops an energetic component that is not described by the thermal core. The boundary between "thermal" and "nonthermal" must be stated. Because the internal energy is a moment of the entire distribution, adding a separately measured tail energy to that same internal energy would double count it.

Takeaway. The ratio of total electron and ion energization is not automatically a thermal heating ratio, and a heating-rate ratio is not the same as a temperature ratio.

5 Physical channels at kinetic scales

The purpose of a mechanism list is to connect a measurable response to a condition under which it can operate.

Landau and transit-time interactions
Parallel electric fields or mirror forces exchange energy with particles whose parallel velocity is near the wave phase speed. The drift-kinetic and quasilinear lectures supply the resonance and velocity-space-diffusion pictures.

Cyclotron interactions
Fluctuations near a gyrofrequency can change perpendicular particle energy. A low-frequency drift-kinetic model averages over the gyrophase and therefore omits this channel.

Stochastic ion heating

Low-frequency fluctuations at $k_\perp \rho_i \sim 1$ can break magnetic-moment conservation when orbit-scale fluctuations become sufficiently strong. A useful control parameter is $\epsilon \sim \delta u_{\rho_i}/v_{\perp i}$; the heating rate is strongly suppressed when ϵ is small [3].

Magnetic pumping

Large-scale compressions and expansions can produce net work when pitch-angle scattering creates the required phase lag between pressure anisotropy and magnetic-field strength.

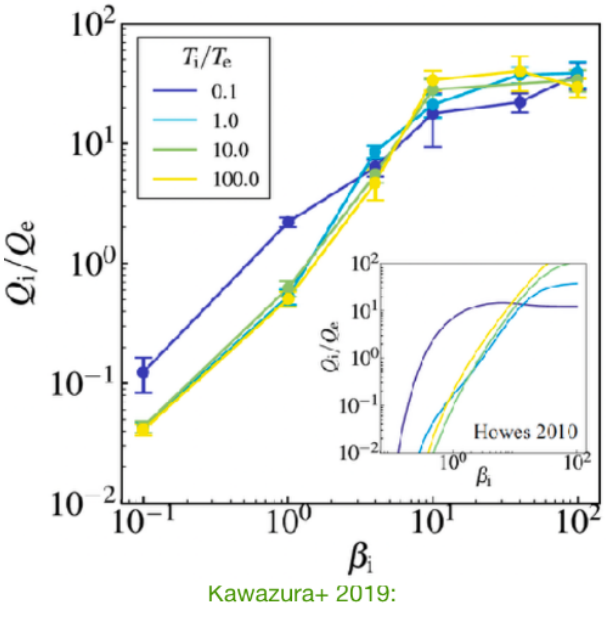
Reconnection and intermittent structures

Turbulence naturally creates current sheets. Reconnection, curvature drift, parallel electric fields, and contracting structures can then energize particles locally. Correlation with current sheets is evidence of where energization occurs, not by itself a complete proof of one universal mechanism.

6 How is power divided between ions and electrons?

The electron–ion partition is not a universal constant. It depends on the kinetic model and on dimensionless parameters such as β_i, temperature ratio, magnetization, fluctuation amplitude, and driving.

For nonrelativistic, low-frequency Alfvénic turbulence, hybrid gyrokinetic calculations find that the ion-to-electron heating ratio increases strongly with β_i and is comparatively insensitive to T_i/T_e over the tested range [4]. At low β_i, electrons receive most of the heat; at high β_i, ions can dominate.



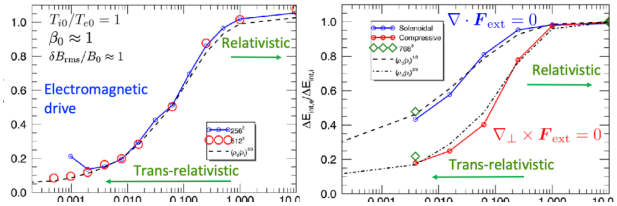


Figure 2: Left: a gyrokinetic ion/electron heating ratio versus β_i [4]. Right: published PIC results illustrating that the electron/ion energization ratio also depends on relativistic temperature and driving [6]. The quantities and model regimes must be checked before the curves are compared.

In relativistic and trans-relativistic electron–ion PIC turbulence, the partition can instead be organized by magnetization, species temperatures, and the ratio of characteristic gyroradii [6]. These simulations include

channels that a gyrokinetic ordering excludes, but generally have less scale separation. Neither result should be promoted to a parameter-independent law.

6.1 Heating rate is not temperature ratio

Even if Q_e/Q_i is known, T_e/T_i also depends on initial conditions, density evolution, the species heat capacities, compression, conduction, radiation, and electron–ion coupling. A closure for Q_e/Q_i belongs inside an evolution equation; it does not directly assign T_e/T_i.

7 Nonthermal acceleration in kinetic turbulence

Kinetic simulations show that turbulence can produce energetic tails as well as heat. The appearance and slope of a tail depend on system size, magnetization, driving, escape, and radiative losses. Relativistic pair-plasma PIC simulations provide a clean example in which a magnetic cascade and a power-law-like particle population develop together [5].

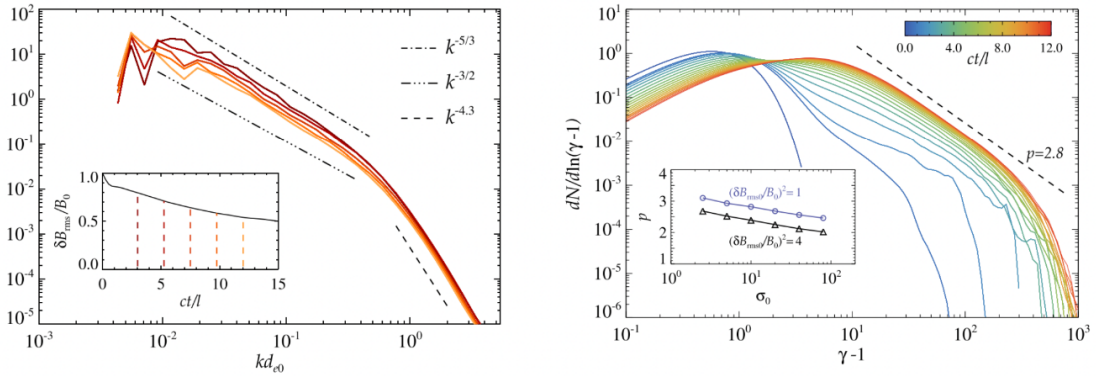


Figure 3: A kinetic-turbulence simulation can be diagnosed simultaneously through its field spectrum and its evolving particle distribution. The two panels answer different questions; the existence of a cascade does not alone fix the nonthermal slope. Figure adapted from published PIC results.

The connection to the previous lecture is the momentum-space Fokker–Planck equation. Tracked-particle increments can define effective coefficients

$$A_p(p) = \lim_{\Delta t\to 0} \frac{\langle \Delta p \mid p\rangle}{\Delta t}, \qquad D_{pp}(p) = \lim_{\Delta t\to 0} \frac{\langle (\Delta p)^2 \mid p\rangle}{2\Delta t},$$

provided that a time interval exists between the correlation time and the macroscopic acceleration time. In relativistic PIC turbulence, measured high-energy coefficients can approach $D_{pp} \propto p^2$, consistent with a second-order-Fermi-like diffusion picture, while a non-negligible drift term remains [7].

This statistical result does not mean that every orbit experiences featureless random kicks. The coefficients may emerge only after averaging over coherent structures and many encounters.

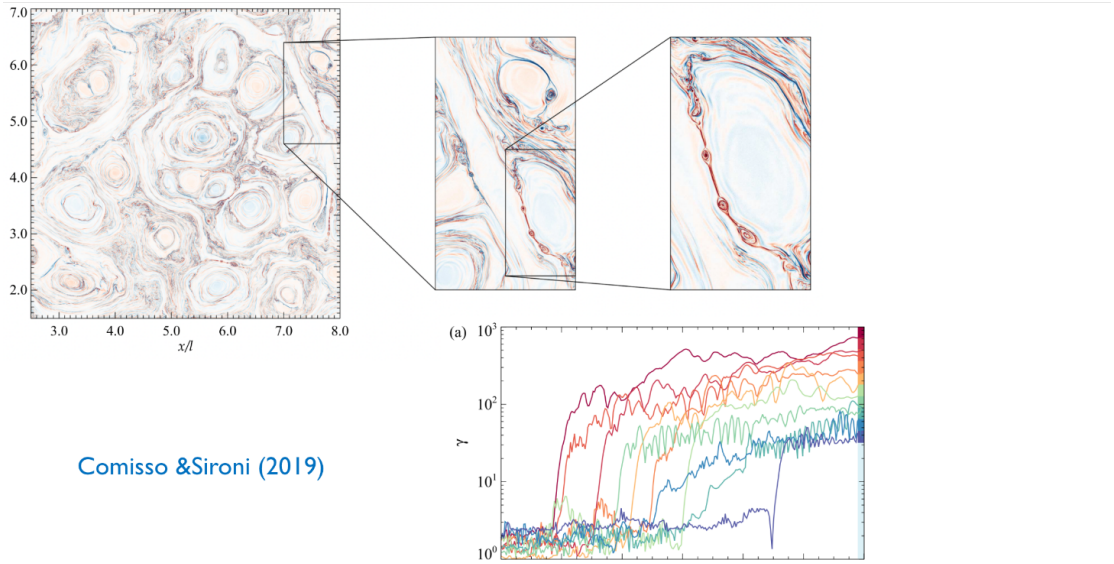


Figure 4: Current sheets and reconnection sites are generated self-consistently by strong turbulence. Particle histories can contain localized episodes of rapid energization embedded in a longer stochastic evolution.

7.1 Intermittent structures and reconnection

At high magnetization, current sheets can be especially important for injection into a nonthermal population. Subsequent stochastic interactions can broaden or extend the tail. The useful conceptual division is therefore not "reconnection or turbulence," because the current sheets are part of the turbulent dynamics. The sharper questions are which structures inject particles, which process controls later diffusion, and how these answers change with plasma parameters.

8 Choosing a kinetic model

No single simulation model covers all scales and mechanisms.

Model	Retains	Typical limitation
Gyrokinetics	Anisotropic low-frequency kinetic response; electron and ion phase-space dynamics	Excludes cyclotron-frequency physics and strong non-gyrotropy by ordering
Hybrid kinetic	Kinetic ions with fluid electrons; ion-scale turbulence and ion heating	Electron kinetic heating and acceleration are modeled rather than resolved
Full PIC	Electron and ion kinetics, reconnection, cyclotron effects, nonthermal acceleration	Restricted mass ratio, box size, duration, or scale separation; numerical noise

Before comparing two simulations, identify the species model, β, magnetization, $\delta B/B_0$, driving, L/ρ_s, dimensionality, and the precise definition of "heating."

9 From kinetic heating to an EHT electron temperature

Low-luminosity black-hole accretion flows illustrate why microscopic partition matters. A GRMHD calculation resolves the fluid-scale dynamics but does not resolve the collisionless kinetic cascade. A sub-grid closure supplies

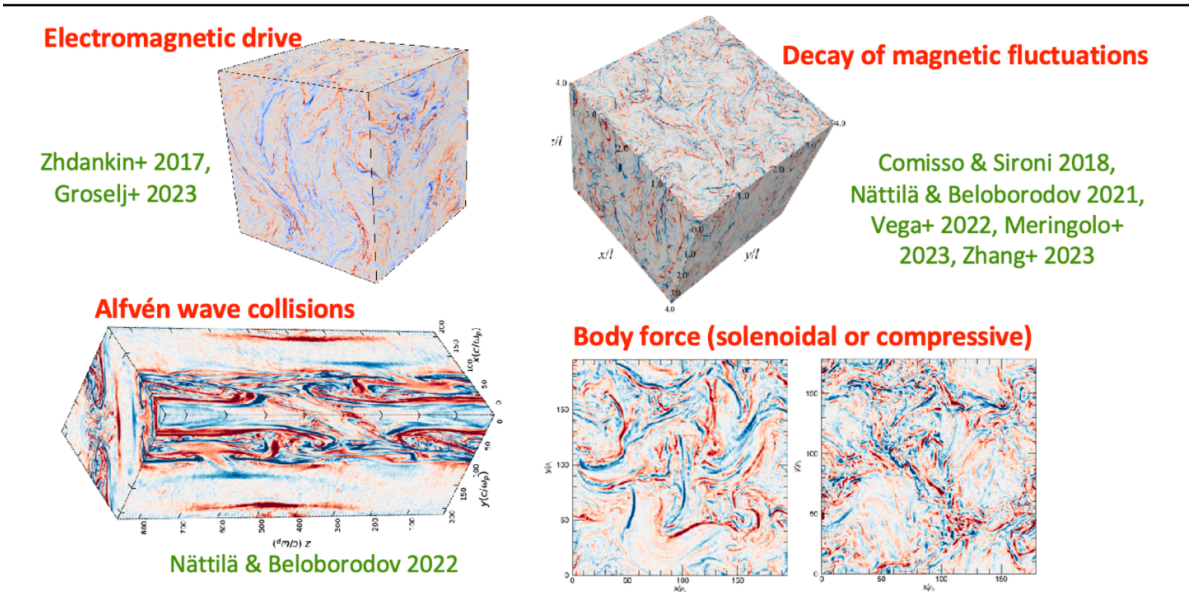


Figure 5: Examples of kinetic-turbulence configurations. Driving method, dimensionality, guide field, species composition, and boundary conditions are part of the physical problem, not merely numerical details.

a fraction

$$\delta_e = \frac{Q_e}{Q_e + Q_i}$$

to an electron internal-energy or entropy equation. In a simple nonrelativistic schematic form,

$$\frac{\partial U_e}{\partial t} + \nabla\cdot(U_e\boldsymbol{u}) = -p_e\nabla\cdot\boldsymbol{u} + \delta_e Q_{\rm diss} - \nabla\cdot\boldsymbol{q}_e - \Lambda_e + Q_{ie}. \tag{4}$$

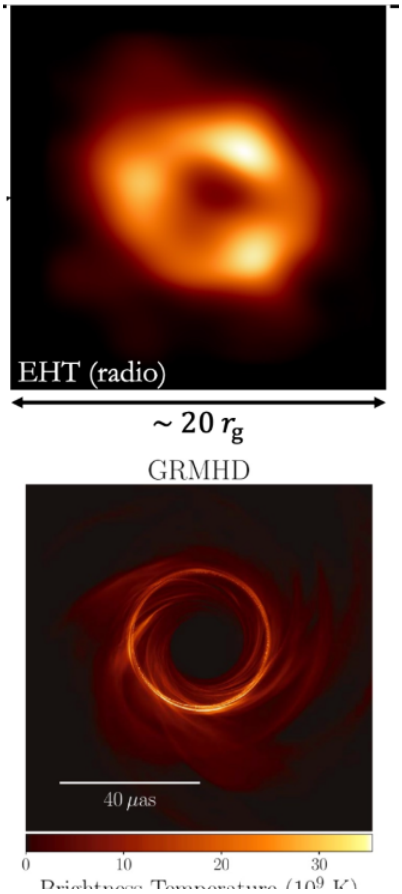
Here Λ_e is radiative cooling and Q_{ie} is interspecies exchange. Relativistic GRMHD uses a covariant version, but the closure enters in the same logical place [8].

An alternative post-processing prescription assigns T_i/T_e directly as a function of plasma beta. That may be useful phenomenologically, but it is logically different from evolving Eq. (4). In particular, $Q_e/Q_i \neq T_e/T_i$ without the intervening thermodynamic history.

10 A compact lecture storyline

The lecture can be delivered with the following progression:

1. Draw the cascade from L through ρ_i to ρ_e and ask where the injected power goes.
2. Write the three energy equations and distinguish $\boldsymbol{j}\cdot\boldsymbol{E}$, pressure–strain conversion, and irreversible heating.
3. Survey the principal kinetic channels without deriving each one.
4. Compare one gyrokinetic and one PIC electron–ion partition result, emphasizing their different assumptions.
5. Use field spectra, particle distributions, and tracked-particle coefficients to discuss nonthermal acceleration.
6. End with the EHT chain: kinetic closure → electron thermodynamics → radiation.



large-scale GRMHD state

$\Downarrow$

kinetic heating closure δ_e

$\Downarrow$

electron thermodynamics and T_e

$\Downarrow$

synchrotron emissivity and image

Figure 6: The kinetic calculation supplies a closure within a longer modeling chain. An observed image does not directly measure a unique microscopic heating ratio.

11 Takeaways

1. Kinetic turbulence transports free energy through both physical and velocity space.
2. $\boldsymbol{j}_s \cdot \boldsymbol{E}$ measures species energization; pressure–strain measures bulk/internal conversion; neither alone proves irreversible heating.
3. Electron–ion partition depends on the plasma regime and on how the heating rate is defined.
4. Thermal heating and nonthermal acceleration are two possible outcomes of the same turbulent energy budget.
5. PIC, hybrid, and gyrokinetic simulations answer different questions because they retain different physics.
6. Macroscopic observables such as EHT images depend on a chain of closures; they do not directly reveal a unique microscopic mechanism.

Recommended reading

- Schekochihin et al. (2009), *Astrophysical gyrokinetics*: the systematic kinetic-cascade framework and the separation of spatial and entropy cascades [1].
- Howes (2010), *A prescription for the turbulent heating of astrophysical plasmas*: a cascade-based heating model and its assumptions [2].
- Kawazura, Barnes, and Schekochihin (2019): electron–ion heating partition in nonrelativistic Alfvénic turbulence [4].
- Zhdankin et al. (2017, 2019): nonthermal acceleration and electron–ion energization in relativistic kinetic turbulence [5, 6].
- Wong et al. (2020): direct measurement of effective Fokker–Planck coefficients from tracked particles [7].
- Ressler et al. (2015): how a kinetic heating prescription enters an electron thermodynamics model for accretion flows [8].

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